

JIAMING HAN

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EDUCATION

Wuhan University (WHU)

Sept.2019 - Present

Master of Photogrammetry and Remote Sensing

GPA: 3.85/4 **IELTS:** 6.5($\forall \geq 5.5$)

Advisor: Prof. Gui-Song Xia

Central South University (CSU)

Sept.2015 - Jun.2019

Bachelor of Remote Sensing Science and Technology

GPA: 85.94/100

Core Courses: Calculus, Linear Algebra, Probability & Statistics, C/C++, Database Theory, Photogrammetry, Remote Sensing, Digital Image Processing, Computer Graphics, Pattern Recognition.

RESEARCH INTERESTS

- **Object detection.** Rotated/3D/Few-Shot/Open-Set Object Detection.
- **Semantic & Instance Segmentation.**
- **Self-supervised Learning.**

RESEARCH PUBLICATION

ReDet: A Rotation-Equivariant Detector for Aerial Object Detection [PDF] [Code(191 ☆)]
IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2021

Jiaming Han, Jian Ding, Nan Xue, Gui-Song Xia

Main idea: We proposed to encode both rotation equivariance and invariance into the detector. The rotation-equivariant backbone extracts features from multiple orientations with single image. Then a rotation-invariant RoIAlign transforms rotation-equivariant features into -invariant features for region-wise object detection in arbitrary oriented aerial images.

Align Deep Features for Oriented Object Detection

[PDF] [Code(214 ☆)]

IEEE Transactions on Geoscience and Remote Sensing (**TGRS**), 2021

Jiaming Han, Jian Ding, Jie Li, Gui-Song Xia

Main idea: We equipped a single stage detector (RetinaNet) with anchor refinement and feature alignment. A novel Alignment Convolution largely reduces the misalignment between anchors and convolutional features. As a result, our method achieves better accuracy *vs.* speed trade-off on several oriented object detection benchmarks.

RESEARCH EXPERIENCE

Tencent YouTu Lab

May.2021 - Present

Research Intern in Computer Vision

Topic: Open-Set Object Detection.

Contributions: (a) Proposed a novel open-set detector (for CVPR2022 submission, in progress) with instance-level contrastive learning (encourages intra-class compactness and inter-class separation) and adversarial unknown identifier (learns compact decision boundary in embedding space).

(b) Implemented an open-set image-review system. For objects of known classes, the detector directly outputs its labels and bboxes, while for objects of unknown (novel) classes, the detector only outputs its bboxes and the labels are identified by a downstream retrieval model.

PROGRAMMING PROJECTS

ReDet: An oriented object detection codebase with >10 sota algorithms and >5 datasets (*e.g.*, sence text and aerial image) support. Code: [ReDet].

WeCSU: A wechat mini-program with a lot of useful information aggregation/querying tools for Central South University. Code: [backend (golang)] and [front-end (javascript)].

SciQuery: An application for searching SCI jounal information. Code: [sciquery].

AWARDS & HONORS

· First Prize of Surveying & Mapping Innovation Competition (CSU)	2018
· Scholarship of China National Geological & Mining	2017
· First-Class Scholarship of Central South University	2017
· First-Class Scholarship of Central South University	2016
· Excellent Student of Central South University	2016

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, PHP, HTML/JavaScript/CSS, Go
- **Deep Learning Frameworks:** PyTorch, TensorFlow, Detectron2/Mmdetection
- **Tools:** Matlab, Linux, Docker, LaTeX